Autonomous Depth-Resolved Sensing of Light, pH, and Temperature at Dana Point

Ben Phung
E80: Experimental Engineering, Harvey Mudd College, Spring 2026

***Abstract*—**Water quality often varies with depth, yet manual sampling at multiple depths is both inefficient and difficult to reproduce consistently. This project focused on designing and deploying a low-cost autonomous underwater vehicle (AUV) capable of collecting depth-dependent measurements of temperature, pH, and light at Dana Point. The motivation for this work comes from local concerns about sewage infrastructure and coastal water quality, which highlights the importance of environmental sensing. The AUV was equipped with an MPX5700 pressure sensor for depth measurement, a Murata NTCNXFT15WB473FA2B150 thermistor for temperature, a glass-electrode pH module for acidity, and two small solar panels to serve as relative light sensors. During the final deployment at Dana Point on April 25, 2026, the vehicle reached a maximum depth of 2.61 ± 0.05 m. Over the course of the descent, the thermistor recorded a temperature decrease from 23.1 ± 0.3 °C above the surface to 18.0 ± 0.6 °C at depth, while the solar-panel channels decreased by 31 ± 5% and 39 ± 10%, respectively. The pH channel remained near the lower end of its calibrated range and did not provide a valid depth-resolved pH profile. The results indicate that the pressure, temperature, and light channels were effective and accurate, while reliable ocean pH sensing will require calibration specific to seawater.

I. INTRODUCTION

The goal of this project was to determine whether a low-cost AUV could provide meaningful water-quality measurements as a function of depth. Temperature and pH are particularly important variables, as ocean warming and acidification can affect stratification, oxygen levels, carbonate chemistry, species distributions, and ecological interactions within marine ecosystems [5], [10]. Light availability, in turn, regulates photosynthesis, primary productivity, and the composition of phytoplankton communities in aquatic environments [11]. Together, these variables are essential for assessing whether surface measurements alone might miss significant environmental changes occurring just a few meters below the surface.

The choice of sensors for this project was guided by four fundamental physical principles. Hydrostatic pressure increases nearly linearly with depth, making it a reliable reference coordinate for the other measured variables. Temperature can vary with depth due to solar heating, mixing, and density-driven thermal gradients. pH is closely connected to the carbonate system, ocean acidification, and biological processes. Light intensity decreases with depth because water and suspended particles absorb and scatter incoming radiation.

While a solar panel does not function as a calibrated oceanographic radiometer, its output can nonetheless serve as a relative indicator of light intensity, provided that factors such as orientation, surface reflection, and wave refraction are recognized as significant sources of error. The solar panels were also chosen because they were extremely inexpensive, helping our team adhere to the strict $50 budget, which had mostly been allocated to purchasing the pH sensor.

The specific objectives of the project were: (1) to calibrate an MPX5700 pressure sensor as a reliable reference for depths between 0 and 3 meters; (2) to determine whether a thermistor could resolve shallow thermal differences; (3) to evaluate whether low-cost solar panels could be used to measure depth-dependent light attenuation; and (4) to assess whether a glass-electrode pH probe could produce a meaningful pH profile in salt water.

II. SENSOR DESIGN

A. Sensor Selection and Calibration Targets

Four analog sensing subsystems connected environmental quantities to Teensy ADC inputs. The MPX5700 pressure sensor was selected as the depth reference because it provides a conditioned analog output over a 0–700 kPa range using a 5 V supply [1]. Over the intended 0–3 m deployment range, hydrostatic pressure changes by only about 29.4 kPa, corresponding to the datasheet's output change from approximately 0.851 V at the surface to 1.040 V at 3 m. Signal conditioning was therefore needed to improve near-surface resolution. The thermistor was a Murata NXFT15WB473FA2B150, specified as 47 kΩ at 25 °C with ±1% resistance tolerance, B = 4050 K, and a 4 s thermal time constant [2]. The light sensors were two ZW-5050-3V solar panels, which have a maximum voltage of 3 V and an open-circuit voltage of 3.3 V [3]. The pH channel used a B08GYF269Y glass electrode interface module with an expected ocean pH range near 7–8 [4].

Prior to deployment, expected sensor output ranges were calculated to ensure that each analog channel could be mapped appropriately to the Teensy ADC range without exceeding the 3.3 V limit. The main governing equations and the final resistor values selected for each channel are summarized in Section II-B. Empirical calibration equations were then applied during data processing for the deployment.

Each sensor channel was calibrated prior to the Dana Point deployment in a controlled environment. The pressure sensor was calibrated in Roberts Pool over a 0–3 m range. The thermistor was calibrated using water baths spanning 10 degrees Celsius to 25 degrees, as we expected the ocean temperature to fall between this range. The pH sensor was calibrated with buffer solutions spanning 7 to 9 pH. The solar panels were calibrated using an iPhone's flashlight feature with controlled flashlight intensities. The resulting calibration fits provided the processing equations used to analyze the deployment dataset.

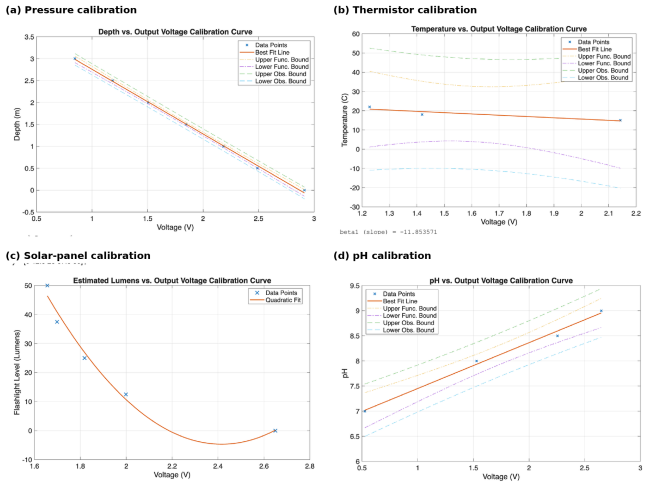


Fig. 1. Sensor calibration curves used to generate the processing equations. Pressure, thermistor, solar-panel, and pH calibrations were performed before deployment using controlled ground-truth measurements.

B. Circuit Design

The signal-conditioning circuits incorporated buffer, attenuation, offset, and gain stages to map the expected sensor outputs into the Teensy 0–3.3 V ADC range. The system was powered by a 12 V battery, with a voltage regulator on the PCB providing a stable 5 V rail for the sensors and MCP601 and MCP-602 op-amps. The Teensy microcontroller then read the conditioned analog outputs. Logged ADC counts were subsequently converted to voltage using the following relation.

$$VADC = (3.3/1023)NADC,$$

Here, NADC represents the Teensy analog reading. For final data processing, empirical calibration curves were used in place of ideal resistor calculations alone, as resistor tolerances, supply variation, and parasitic effects from the soldered board altered the actual transfer functions.

For each analog channel, the required span gain or attenuation was selected from the expected input-voltage span and desired output-voltage span:

$$Gspan = (Vout,max - Vout,min)/(Vin,max - Vin,min).$$

Reference offsets were produced with voltage dividers [13],

$$Vref = 5Rg/(Rp + Rg),$$

and the offset/gain stages were modeled with the op-amp relation [13]

$$Vout = (1 + Rf/Rn)Vin - (Rf/Rn)Vref.$$

For any purely inverting stage, the gain relation was also taken from the standard inverting-amplifier model [13]

$$Vout = -(Rf/Rin)Vin.$$

These equations guided the selection of resistor ratios to maximize use of the ADC range while ensuring that all expected outputs remained below 3.3 V.

The pressure channel used the hydrostatic pressure relation for a constant-density fluid [12] and the MPX5700 transfer function from the sensor datasheet [1] to estimate the raw deployment signal:

$$P = P0 + \rho gh$$

$$VP = Vs(0.0012858P + 0.04).$$

Over 0-3 m, the raw pressure-sensor output was expected to span approximately 0.851-1.040 V. This span was mapped toward 0.5-2.7 V using the offset/gain architecture shown in Fig. 3. The final pressure interface used approximately Rn = 1 kΩ, Rf = 11 kΩ, and a 10 kΩ/40 kΩ reference divider before empirical calibration. Roberts Pool calibration produced

$$h = -1.4708V + 3.9236,$$

where h is depth in meters and V is conditioned pressure voltage.

The thermistor channel used the Murata beta model and divider equation,

$$R(T) = R25\ exp[B(1/T - 1/T25)]$$

$$Vdiv = 5Rb/(Rth + Rb).$$

Using R25 = 47 kΩ, B = 4050 K, and Rb = 47 kΩ, the expected 10-25 °C ocean range produced a divider output of about 1.65-2.50 V. A unity-gain buffer prevented the high-impedance divider from being loaded by the offset amplifier. The offset stage used Rn = 120 kω, Rf = 270 kω, and a 17 kω/10 kω reference divider. The final thermistor processing equation was

$$T = -11.854V + 32.424.$$

The pH channel used a glass-electrode interface module. Buffer testing gave an expected module output of approximately 0.7-1.2 V over the calibration range, so the span was expanded using Rf = 44 kΩ, Rn = 10 kΩ, and a 37 kΩ/10 kΩ reference divider. The buffer calibration produced

$$pH = 0.912V + 6.538.$$

The solar-panel channels primarily required attenuation because bright-day tests showed that the raw outputs could approach the Teensy's limit. The expected 0.5-3.6 V panel range was compressed toward 0.5-2.7 V using Rf = 71 kΩ, Rn = 100 kΩ, and an 18 kΩ/10 kΩ reference divider. The solar calibration used

$$I = 88.58V2 - 427.97V + 512.27,$$

In this equation, I represents the estimated light intensity in lumens. To allow for adjustments in the field, 1×1 female headers were soldered to the PCB, allowing resistor values to be swapped between planned 3 m, 5 m, and 9 m configurations without desoldering components.

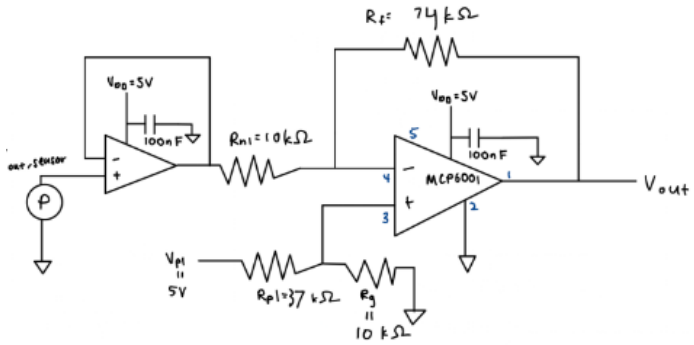


Fig. 2. Thermistor circuit design using a voltage divider, unity gain buffer, and offset to convert temperature-dependent resistance into a conditioned voltage.

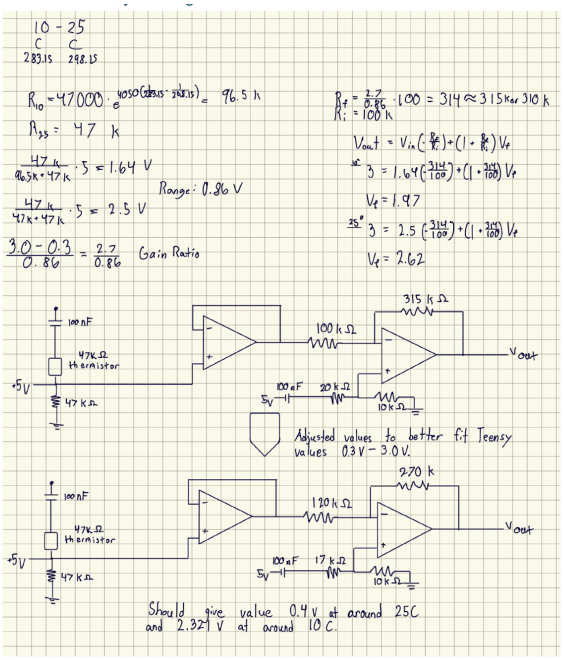


Fig. 3. Thermistor circuit design using a voltage divider, unity gain buffer, and offset to convert temperature-dependent resistance into a conditioned voltage.

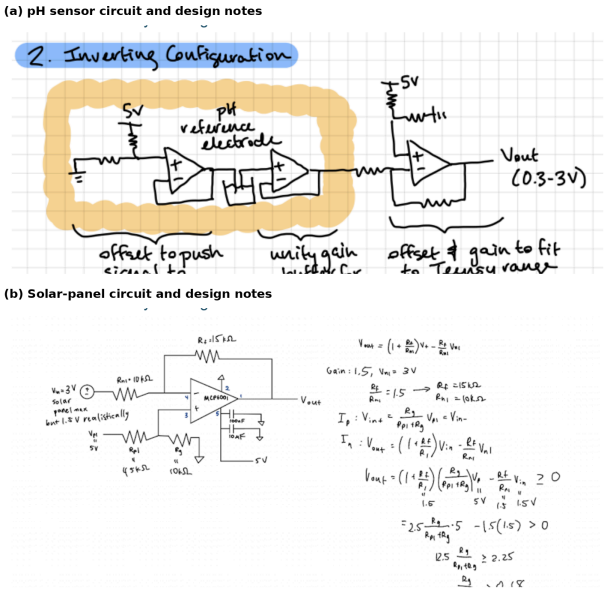


Fig. 4. pH and solar-panel circuit designs. Both circuits used offset/gain stages to map expected sensor voltages into the Teensy ADC range.

III. ROBOT DESIGN

The final AUV design featured a compact PVC frame, prioritizing vertical movement. The pressure sensor, thermistor, and pH probe were mounted facing downward so that their sensing surfaces would contact water at approximately the same depth. The solar panels were oriented upward to measure incident light during descent. The waterproof electronics enclosure was positioned near the center of the frame, and buoyant elements were placed above heavier components to enhance passive stability. Finally, the robot's frame was altered to be cubic, with four vertically pointing motors at each corner.

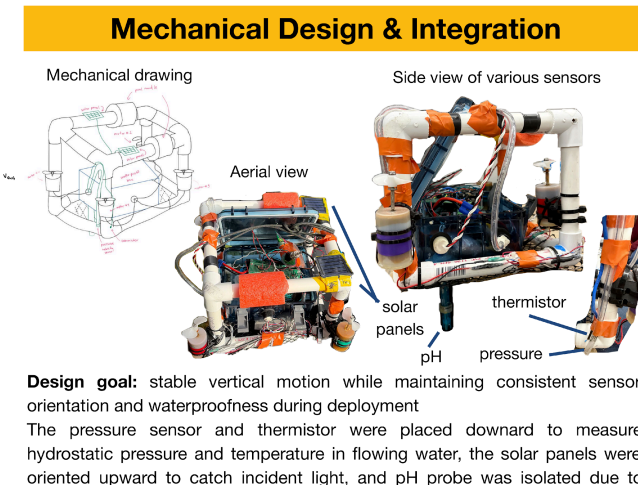


Fig. 5. Final mechanical integration from the project poster. The final build used upward-facing solar panels and downward-facing pressure, pH, and thermistor sensors to maintain consistent orientation during vertical movement.

The AUV's mass was estimated at 3.3701 kg, corresponding to 33.06 N. The calculated lifting force was 39.43 N, resulting in 6.37 N excess buoyancy, which would require approximately 0.649 kg of ballast to achieve neutral buoyancy. The center of mass was estimated to be 0.116 m above the bottom of the frame, while the center of buoyancy was 0.135 m above the bottom, making sure the buoyancy center was above the mass center and providing a restoring moment after small rotations. The initial motor configuration used four vertical motors at the corners, but two of them drew excessive current, causing reliability issues with the motherboard. The final design employed two diagonal vertical motors to improve electrical reliability.

Roberts Pool and pHake Lake served as freshwater testing sites, offering conditions distinct from those at the final Dana Point saltwater deployment. Roberts Pool provided a sufficiently controlled environment for pressure calibration over the 0–3 m range. However, the initial deployment at pHake Lake revealed a significant waterproofing failure, which resulted in the electronics box being flooded, the motor-controller warning lights activated, and most sensor channels returning flat, railed, or shorted signals. As a result, the Dana Point deployment placed greater emphasis on thorough waterproofing checks, immediate tethered recovery, and a staged depth plan to ensure that usable data could be collected before attempting deeper dives.

The mechanical integration required waterproofing because the electronics box had multiple penetrations for motors and sensors that would be on the robot's exterior. Penetrator holes were drilled with a step drill to create cleaner circular openings, sensor and motor wires were routed through sealed penetrators, and marine epoxy was applied around cable paths to waterproof the holes. The enclosure was checked before electronics were installed to detect leaks without risking the motherboard or battery. This process became more important after the first pHake Lake failure, where water entered the box and caused our circuits to short.

The pH circuit required additional electrical isolation because the glass-electrode measurement is much more sensitive to noise than the pressure, thermistor, and solar-panel channels.

The pH probe produces a small electrochemical signal through a high-impedance interface, so nearby motor wires, PWM switching, long sensor leads, and motion of the tethered robot could introduce voltage fluctuations before the signal reached the Teensy. To reduce this effect, the pH interface circuit was made on a separate breadboard and placed inside a grounded Faraday cage made of copper and aluminum tape. This shielding helped block external electric fields and made the pH channel less vulnerable to noise from the robot's motors and other onboard electronics.

IV. DEPLOYMENT DETAILS

The main deployment dataset was collected at Dana Point on April 25, 2026. Pre-deployment preparations began at 7 a.m., with deployment starting at 8 a.m. and testing continuing until about 1 p.m. The analyzed dataset was recorded around 11 a.m. Morning conditions were cloudy, with sunlight increasing throughout the day and reaching full sun by approximately 1 p.m.

Prior to launch, analog outputs were checked using the Arduino IDE serial monitor to confirm that the pressure, thermistor, pH, and solar-panel channels were reaching the Teensy and producing plausible values. The waterproof enclosure was sealed and inspected, wires were cleared from the propellers, and the delayed motor start was verified before launching the robot from the dock shown in Fig. 6. Most tests were conducted from the dock and lower stair platform to keep laptops and tools accessible for debugging, while several deployments used a two-person kayak to position the robot farther from the dock. Safety protocols included the use of life jackets during kayak operations, a 10 m tether for robot retrieval, and immediate recovery after each run. If the robot sank, lost power, or began to flood, the tether served as the primary recovery method. Tape labels placed every 0.5 m on the tether provided a visual depth reference during deployment.

Originally, the deployment plan included a 9 m dive. Because different target depths required different resistor values, individual 1×1 female headers were soldered onto the PCB in place of fixed resistors so that values could be swapped without desoldering components in the field. This allowed us to use the full range of the Teensy's reading capabilities. At Dana Point, however, early complications led us to switch back to a 3 m dive only rather than continue toward the 9 m goal. MATLAB calibration scripts converted logged ADC counts to volts and subsequently to depth. The descent interval selected for analysis was identified from the pressure trace, ensuring that temperature and light comparisons were referenced to the same pressure-derived depth coordinate.

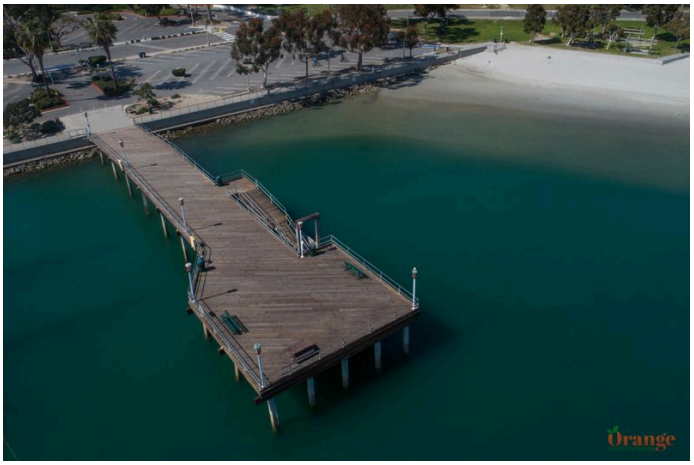


Fig. 5. Aerial view of the dock and shore where robot reployments were made.

V. RESULTS AND ANALYSIS

A. Pressure-sensor data

The pressure sensor provided the most reliable quantitative channel. Fig. 7 shows that the AUV remained near the surface, descended to a maximum calibrated depth of 2.61 ± 0.05 m, held near its deepest interval, and returned to the surface. The uncertainty estimate uses the Roberts Pool pressure calibration RMS residual of 0.0462 m, rounded to ±0.05 m. Because this residual is small relative to the 2.61 m range, pressure provided a sufficient independent coordinate for comparing trends in temperature, light, and pH.

Additional systematic error can arise from the pressure sensor port depth differing slightly from the frame reference point. The pressure result remains the most defensible coordinate because the sensor response time is about 1 ms, which is much faster than the vehicle's motion. Fig. 7 is plotted as a continuous time series for readability; uncertainty bars are summarized in the depth-related profile in Fig. 8.

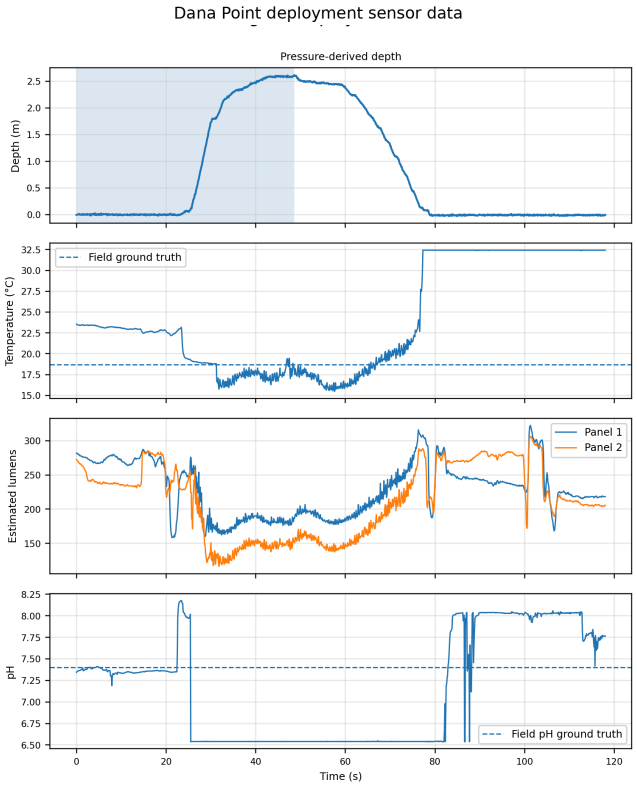


Fig. 7. Dana Point deployment sensor time series. The pressure channel provides the depth reference; thermistor and solar signals change during descent; the pH channel clips near the low end of its calibrated range.

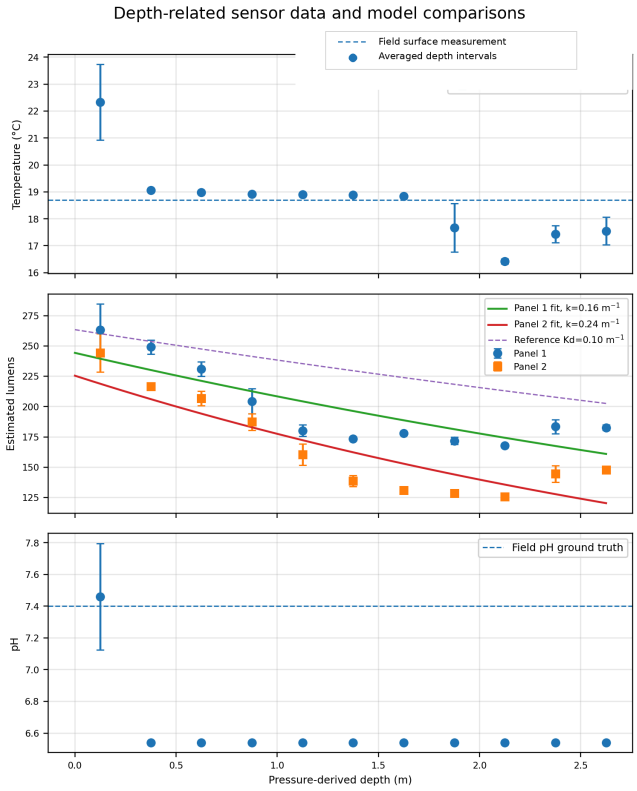


Fig. 8: Depth-related sensor results with uncertainty estimates. Solar panel fits are compared with an exponential attenuation reference curve (Kd = 0.10 m⁻¹), and pH data show clipping rather than a physically valid profile.

B. Temperature-sensor data

The thermistor data showed a clear cooling trend during the dive. The pre-dive interval averaged 23.1 ± 0.3 °C, while the deepest interval averaged 18.0 ± 0.6 °C. However, an independent field thermometer measured the surface water temperature at 18.7 °C. This discrepancy indicates that the thermistor was more effective at capturing relative temperature trends than providing an absolute calibrated temperature. Possible sources of error include intercept error, insufficient equilibration before launch, heat from nearby electronics, or thermal lag during motion.

Therefore, the thermistor data should be interpreted as reflecting relative temperature trends rather than as providing calibrated absolute temperature measurements. Given the thermistor's approximate 4-second time constant, making repeated fixed-depth holds necessary to distinguish actual water-column structure from sensor lag.

C. Solar-Panel Light Attenuation

Both solar-panel channels showed decreased output with increasing depth, consistent with the expected pattern of light attenuation. The pre-dive surface interval averaged 273 ± 6 estimated lumens for solar panel 1 and 252 ± 19 estimated lumens for solar panel 2. At the deepest interval, the averages were 191 ± 8 and 157 ± 7 estimated lumens, respectively. These changes correspond to 30 ± 3% attenuation for panel 1 and 38 ± 6% attenuation for panel 2 over the course of the descent.

Depth-related exponential fits yielded $k = 0.16 \pm 0.08$ m^{-1} for solar panel 1 and $k = 0.24 \pm 0.11$ m^{-1} for solar panel 2, with R^2 values of 0.68 and 0.72. These coefficients correspond to e-folding lengths of approximately 6.3 m and 4.2 m. NOAA identifies Kd490 = 0.1 m^{-1} as an attenuation coefficient representing a 10 m e-folding depth in blue-green light, with higher Kd values indicating reduced water clarity [7]. The measured coefficients are physically plausible for shallow coastal waters and a broadband solar-panel response; however, these values should not be considered site-quality optical measurements due to the influence of tilt, surface-wave focusing, frame shadowing, and broadband spectral sensitivity on the panels.

The differences observed between the two solar-panel channels can be attributed to factors such as orientation, frame shading, surface ripples, bubbles, and the panels' broadband spectral response. As a result, the solar panels were used as relative light sensors rather than as calibrated instruments for optical water quality.

D. pH Result and Salinity Error

The pH channel did not yield a valid depth-resolved pH measurement. During the descent, the calibrated output converged near pH 6.54 and remained nearly constant, even though the field ground truth was approximately pH 7.4. This result suggests signal clipping or a loss of usable electrode response, rather than a true chemical gradient. The buffer calibration residual of 0.0899 pH units does not reflect the deployment error, since the chemistry of the deployment environment differed substantially from that of the calibration solutions.

This result corresponds to established limitations of glass electrodes in seawater. The course salinity correction note indicates that ocean water alters the liquid-junction potential and the ionic environment relative to low-salinity buffers, estimating that ocean pH near 8 could appear approximately 0.5 pH units lower under the class conditions [9]. Martell-Bonet and Byrne also report that glass electrodes must be calibrated in solutions matching the sample salinity, and that neglecting salinity dependence can result in pH errors up to 0.24 pH units within their study range [8]. Therefore, the pH result was not attributable solely to wiring failure or a circuit design issue. It demonstrated that low-cost ocean pH sensing requires seawater-specific calibration, high-input-impedance buffering, shielding, and extended settling time.

E. Model Agreement and Limitations

The strongest agreement with the expected model was found in the pressure channel, where the calibrated depth profile matched the anticipated descend-hold-return motion and showed a residual of only 0.0462 m. The temperature and light channels exhibited meaningful trends, but lacked robust absolute validation. The thermistor trend qualitatively matched the expected variation of temperature with depth, although the surface ground truth indicated an offset error. The solar-panel data qualitatively followed the pattern of exponential light attenuation and produced k values similar to the 0.1 m^{-1} reference; however, the fit quality and sensor characteristics do not support quantitative conclusions about water clarity. The pH channel did not agree with the ground truth and was therefore excluded. The uncertainty estimates provided here should be regarded as engineering approximations rather than as comprehensive metrological uncertainty budgets. Pressure uncertainty was based on the Roberts Pool calibration residual, which captures much of the voltage-to-depth calibration scatter but does not account for all possible systematic shifts, such as differences in water density between pool water and ocean water or minor atmospheric pressure variations between the pool and harbor. Temperature uncertainty was estimated from the observed scatter within selected depth ranges, however, the larger absolute discrepancy with the 18.7 °C field thermometer suggests that calibration offset was the primary source of error. Solar-panel uncertainty bars were calculated from the variation across the selected depth ranges in the logged signal and mainly reflect short-term sensor scatter and wave-induced

The main experimental limitation is that the deployment dataset consists only of a single successful profile, rather than multiple repeated dives. While a single dive can reveal depth-dependent trends, it does not allow for separation of environmental variation from elements such as robot motion, sensor orientation or changing sunlight. For example, both solar channels decreased with depth, but surface ripples and vehicle tilt can focus or shadow light on the panels over time scales similar to descent. Similarly, the thermistor cooling trend is consistent with a real water-column temperature difference, but without fixed depth holds or repeated descents, it is difficult to distinguish the sensor time constant from the actual water profile.

Despite these limitations, the agreement between the qualitative models and the measured trends remains meaningful. The pressure trace followed the intended descend-hold-return behavior, the thermistor readings approached the independent field temperature during immersion, and both solar-panel channels decreased during the deepest part of the profile. The pH channel was the exception, as it did not agree with ground truth or the expected seawater pH range, and was therefore treated as a diagnostic result regarding sensor suitability rather than as environmental data.

VI. CONCLUSION

The Dana Point deployment demonstrated that a small AUV is capable of detecting depth variation in a shallow coastal water column over a descent of 2.61 ± 0.05 m. The collected data describe a shallow profile characterized by measurable cooling and reduced light availability with increasing depth, both of which are pertinent to marine habitat conditions, as temperature influences organism physiology and stratification, while light constrains photosynthesis and primary production. The observed light-attenuation coefficients and the 31–39% decrease in solar-panel output indicate detectable optical structure in the deployment area, although the solar panels should be regarded as relative light sensors rather than calibrated radiometers.

The pH result demonstrated the primary design lesson. A glass electrode calibrated in low-salinity buffers and then deployed in seawater did not yield a credible pH-depth profile. Future iterations should therefore use calibration solutions with seawater-like salinity, shield the high-impedance probe, and allow longer settling times before logging pH. Operationally, the use of a tether, serial-monitor checks, and a staged-depth strategy should be maintained to support recoverable failures. For future circuit boards, sealed, modular gain settings should be implemented instead of exposed, removable headers for field adjustment. The most important remaining improvement is experimental repeatability. Subsequent deployments should include repeated, controlled descents with 20–30 second holds at each depth to distinguish temperature and light trends from wave focusing, sensor orientation, and transient motion.

ACKNOWLEDGEMENTS

Thank you, Xavier Walter, for help with debugging circuits and electronics, Lynn Kim for help with the pH sensor, Faraday cage, and lab forensics on the pH failure, Georgia Tai for circuit debugging, conceptual support, and persistence during difficult debugging periods, Joshua Ikehara for circuit debugging, prelab help, and project guidance; and Tracy Han for prelab discussions and advice. The HMC stockroom provided key sensing components, including the pressure sensor, thermistor, and resistor values. The Machine Shop and Makerspace supported the PVC frame, waterproofing strategy, and tether fabrication. Prof. Yang, the E80 teaching staff, and the E80 TAs provided course support throughout design, testing, and deployment. I would also like to give a special thank-you to my team members, Luke Pratt, Megan Tran, and Lucy Will, for joining me on this incredible journey and making E80 so much more fun!

REFERENCES

[1] NXP Semiconductors, “MPX5700 Integrated Silicon Pressure Sensor: On-Chip Signal Conditioned, Temperature Compensated and Calibrated,” data sheet, Rev. 10, Oct. 2012.

[2] Murata Manufacturing Co., Ltd., “NXFT15WB473FA2B150 Product Search Data Sheet,” Feb. 2024.

[3] Zhiwang/Jameco ValuePro, “ZW-5050-3V Mini Solar Panel,” product data sheet.

[4] HNNUY, “Liquid pH 0-14 Value Detect Sensor Module,” product manual, B08GYF269Y.

[5] NOAA Ocean Acidification Program, “Ocean Acidification Monitoring and Modeling,” National Oceanic and Atmospheric Administration.

[6] C. J. Weiskerger, J. A. Rowe, J. Stow, and C. M. Stuart, “Application of the Beer-Lambert model to attenuation of photosynthetically active radiation in a shallow, eutrophic lake,” Water Resources Research, vol. 54, no. 11, pp. 8952–8962, 2018.

[7] NOAA Fisheries InPort, “Light Attenuation Kd490,” National Marine Fisheries Service.

[8] L. Martell-Bonet and R. H. Byrne, “Characterization of the nonlinear salinity dependence of glass pH electrodes: A simplified spectrophotometric calibration procedure for potentiometric seawater pH measurements at 25 °C in marine and brackish waters: $0.5 \leq S \leq 36$,” Marine Chemistry, vol. 220, 103764, 2020.

[9] E80 Teaching Team, “pH Measurement Correction for Salinity,” course note, Apr. 29, 2026.

[10] S. C. Doney et al., “Climate change impacts on marine ecosystems,” Annual Review of Marine Science, vol. 4, pp. 11–37, 2012, doi: 10.1146/annurev-marine-041911-111611.

[11] Q. Lin et al., “Light attenuation parameterization in a highly turbid mega estuary and its impact on the coastal planktonic ecosystem,” Frontiers in Marine Science, vol. 11, 2024, doi: 10.3389/fmars.2024.1486261.

[12] OpenStax, “14.1 Fluids, Density, and Pressure,” University Physics Volume 1. Rice University, 2016. [Online]. Available: https://openstax.org/books/university-physics-volume-1/pages/14-1-fluids-density-and-pressure.

[13] R. Mancini, Ed., “Op Amps for Everyone: Design Guide,” Texas Instruments, SLOD006B, Aug. 2002. [Online]. Available: https://www.ti.com/lit/an/slod006b/slod006b.pdf.

[14] Orange County Outdoors. *Dana Point Pier*. Retrieved May 11, 2026, https://www.orangecountyoutdoors.com/get-out-there/piers/dana-point-pier